

IZYUM, Ukraine

WMO: 344150

Lat: 49.1872N

Lon: 37.2894E

Elev: 78

StdP: 100.39

Time Zone: 2.00 (EUE)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-21.5	-17.6	-24.1	0.4	-21.2	-20.7	0.6	-17.3	7.7	-3.2	6.7	-3.0	1.1	340	0.478

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	12.4	33.7	20.5	31.7	19.9	30.1	19.5	22.1	29.7	21.3	28.8	20.5	27.8	2.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	14.6	24.9	18.9	13.8	24.1	18.1	13.2	23.2	65.1	29.8	62.1	28.9	59.5	28.0	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
7.5	6.1	5.1	-26.0	36.1	3.7	2.3	-28.6	37.8	-30.8	39.1	-32.9	40.4	-35.5	42.1		
			-26.1	24.0	3.6	1.5	-28.8	25.1	-30.9	25.9	-32.9	26.8	-35.6	27.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-4.5	-3.4	2.0	9.9	16.0	20.0	22.1	20.8	14.9	8.2	2.6	-2.3
	DBStd	10.56	6.43	6.45	4.55	4.01	4.25	3.52	3.03	3.49	4.14	4.62	5.00	5.97
	HDD10.0	1837	450	375	251	50	5	0	0	0	8	91	227	381
	HDD18.3	3763	708	608	508	255	97	22	4	16	117	316	473	639
	CDD10.0	1438	0	0	1	46	191	301	374	334	154	34	4	0
	CDD18.3	322	0	0	0	1	24	72	120	91	13	1	0	0
	CDH23.3	3814	0	0	0	17	343	793	1302	1147	203	8	0	0
	CDH26.7	1443	0	0	0	2	98	271	528	493	51	1	0	0
Wind	WSAvg	2.0	2.1	2.3	2.5	2.2	1.9	1.8	1.6	1.5	1.6	1.8	2.1	2.2
Precipitation	PrecAvg	569	45	39	44	41	50	73	49	39	51	44	41	51
	PrecMax	955	84	109	87	123	129	349	154	149	107	80	121	119
	PrecMin	400	15	11	9	4	6	12	4	4	0	9	1	18
	PrecStd	122	20	22	22	28	29	63	33	29	34	20	29	28
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	7.8	10.3	17.0	25.5	32.2	33.5	36.2	36.5	30.9	24.1	16.6	10.5
		MCWB	6.5	8.0	10.4	14.0	18.5	20.5	20.8	20.7	18.5	16.1	12.8	9.0
	2%	DB	5.6	6.9	13.5	22.4	29.3	31.5	33.8	33.9	28.1	20.7	13.8	8.1
		MCWB	4.7	5.2	7.8	13.0	17.1	19.9	20.9	20.2	17.1	14.4	11.4	7.0
	5%	DB	4.1	5.0	11.1	20.3	27.2	29.8	31.6	31.6	25.7	17.6	11.4	6.3
		MCWB	3.3	3.7	6.8	12.0	16.9	19.6	20.4	19.6	16.2	13.5	9.6	5.4
	10%	DB	2.6	3.6	8.6	17.9	24.8	27.9	29.7	29.5	23.2	15.3	9.3	4.6
		MCWB	1.9	2.6	5.7	10.7	16.1	18.8	20.1	19.2	15.6	12.4	7.9	3.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.0	8.4	10.6	15.3	20.3	22.6	23.3	22.8	20.2	17.1	13.3	9.2
		MCDB	7.8	10.1	15.8	23.3	28.8	29.9	30.8	31.0	28.2	20.8	15.3	10.1
	2%	WB	4.8	5.5	8.7	14.0	18.8	21.5	22.2	21.7	18.6	15.4	11.6	7.1
		MCDB	5.4	6.6	12.0	20.4	26.0	28.6	29.6	30.0	25.2	19.0	13.5	8.1
	5%	WB	3.3	3.9	7.2	12.9	17.9	20.6	21.5	20.7	17.3	13.9	9.7	5.5
		MCDB	4.0	4.8	10.1	18.1	24.3	27.3	28.8	28.8	22.9	17.0	11.1	6.2
	10%	WB	1.9	2.6	5.9	11.8	16.9	19.6	20.8	19.9	16.3	12.5	8.0	3.9
		MCDB	2.4	3.5	8.5	16.5	23.0	26.0	27.9	27.3	21.5	15.1	9.0	4.6
Mean Daily Temperature Range	5% DB	MDBR	5.4	6.3	7.7	11.2	12.5	11.9	12.4	13.9	12.7	9.5	5.6	4.9
		MCDBR	4.2	6.8	12.5	15.8	16.1	14.9	15.5	17.6	17.0	14.2	8.1	5.1
		MCWBR	3.9	5.4	7.5	7.5	6.0	5.3	5.0	5.8	7.4	8.1	5.9	4.6
	5% WB	MCDBR	4.1	6.2	10.5	12.8	13.1	12.6	13.0	14.6	14.0	11.8	7.1	5.0
MCWBR		3.8	5.2	7.0	6.5	5.7	5.2	5.2	5.8	7.1	7.3	5.6	4.5	
Clear-Sky Solar Irradiance	taub	0.328	0.358	0.390	0.462	0.419	0.413	0.443	0.441	0.411	0.375	0.355	0.332	
	taud	2.153	2.081	2.143	2.043	2.258	2.301	2.224	2.216	2.265	2.326	2.284	2.221	
	Ebn at Noon	687	756	805	782	842	849	815	796	780	743	661	630	
	Edh at Noon	82	112	126	155	130	125	134	129	111	88	74	69	
All-Sky Solar Radiation	RadAvg	0.95	1.88	3.01	4.31	5.75	6.15	6.08	5.43	3.74	2.15	0.95	0.60	
	RadStd	0.19	0.36	0.43	0.43	0.45	0.51	0.41	0.40	0.43	0.35	0.19	0.10	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.69	N/A	N/A	N/A	N/A	N/A	-158	-210	+86	+65
(47)	Regional (6 neighbors)	+0.80	N/A	N/A	+1.12	N/A	N/A	-153	-223	+130	+77

Nomenclature: See separate page